Software Engineering (CSC702)

Assignment 4

Online Shopping System

Software Requirements Specification, Software Design Document,
Software Test Plan and Test Report

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1 Software Requirements Specification (SRS)

1.1 Introduction

Purpose. This document specifies the requirements of the Online Shopping System, a web-based e-commerce application. It is intended for developers, testers, the course instructor and any stakeholder who needs to understand what the system must do.

Definitions and abbreviations.

Term	Meaning
SRS	Software Requirements Specification
SDD	Software Design Document
UI	User Interface
COD	Cash on Delivery
UPI	Unified Payments Interface
SKU	Stock Keeping Unit
Payment Gateway	External service that processes online payments

References. IEEE Std 830-1998 and ISO/IEC/IEEE 29148 (SRS); IEEE Std 1016 (SDD); IEEE Std 829 (test documentation).

1.2 Problem Statement

Customers who buy from physical stores face limited product choice, restricted opening hours, difficulty comparing prices and no easy way to follow a purchase after paying. Small and medium sellers also lack a simple way to list products, manage stock and handle orders. The Online Shopping System is required to let customers browse products, add items to a cart, place orders, pay online and track deliveries, while giving administrators the tools to manage products, stock, orders and users.

1.3 Objectives

- Provide a simple, secure platform for browsing and purchasing products online.
- Allow customers to search and filter products and view full product details.
- Support a shopping cart, checkout, multiple payment methods and order confirmation.
- Let customers track the status of every order until delivery.
- Give administrators control over the catalogue, inventory, orders and user accounts.
- Keep customer and payment data safe and the system responsive and available.

1.4 Scope of the System

In scope: user registration and authentication; product catalogue with categories, search and filters; shopping cart; checkout and order placement; online payment through an external gateway and COD; order tracking and history; order cancellation before shipping; ratings and reviews; admin management of products, inventory, orders and users; basic sales reports.

Out of scope: physical logistics of courier companies, multi-vendor seller onboarding, native mobile applications, return and refund workflows beyond cancellation, and recommendation engines.

1.5 Functional Requirements

ID	Requirement	Priority
FR1	A visitor shall be able to register with name, email, phone and password.	High
FR2	A registered user shall be able to log in and log out securely.	High
FR3	A customer shall be able to view and edit profile details and saved addresses.	Medium
FR4	The system shall display products grouped by category.	High
FR5	The system shall allow searching by keyword and filtering by category, price range and rating.	High
FR6	The system shall show product details: name, images, description, price, stock status and reviews.	High
FR7	A customer shall be able to add a product to the cart.	High
FR8	A customer shall be able to change quantity or remove items in the cart; the total shall update automatically.	High
FR9	A customer shall be able to select or enter a delivery address at checkout.	High
FR10	A customer shall be able to place an order from the cart.	High
FR11	The system shall support payment by card, UPI, net banking and Cash on Delivery.	High
FR12	The system shall confirm payment success or failure and update the order accordingly.	High
FR13	The system shall send an order confirmation to the customer by email.	Medium
FR14	A customer shall be able to track the current status of an order.	High
FR15	A customer shall be able to view past orders.	Medium
FR16	A customer shall be able to cancel an order before it is shipped.	Medium
FR17	A customer shall be able to rate and review products they have purchased.	Low
FR18	An administrator shall be able to add, edit and delete products and categories.	High
FR19	An administrator shall be able to update stock levels; the system shall reduce stock when an order is placed.	High
FR20	An administrator shall be able to view orders and update their status (Packed, Shipped, Out for Delivery, Delivered).	High
FR21	An administrator shall be able to view, block or unblock user accounts.	Medium
FR22	An administrator shall be able to view sales reports.	Low

1.6 Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	Pages shall load within 3 seconds under normal load; the system shall support at least 500 concurrent users.
NFR2	Security	All traffic shall use HTTPS; passwords shall be stored hashed; card details shall never be stored by the system and shall be handled only by the payment gateway.
NFR3	Reliability	The system shall have at least 99% uptime and shall not lose confirmed orders.

ID	Category	Requirement
NFR4	Usability	A new user shall be able to place an order without training; the UI shall be responsive on desktop and mobile browsers.
NFR5	Scalability	The design shall allow adding servers and products without re-design.
NFR6	Maintainability	The code shall be modular and documented so that a feature can be changed with limited impact on others.
NFR7	Compatibility	The system shall work on recent versions of Chrome, Firefox, Edge and Safari.
NFR8	Data integrity	Stock counts and order totals shall remain consistent when several users order the same product at once.
NFR9	Backup	Data shall be backed up daily.

1.7 User Requirements

User	Needs
Customer	Register and log in; search and browse products; manage cart; place and pay for orders; track and cancel orders; review products.
Administrator	Manage catalogue and stock; process orders and update delivery status; manage users; view reports.
Payment Gateway (external system)	Receive payment requests from the system and return success or failure responses.

1.8 System Requirements

- The system shall be a three-tier web application: presentation, application logic and data storage.
- The system shall expose its functions through a web interface accessible from a browser.
- The system shall integrate with an external payment gateway through its API.
- The system shall use a relational or document database to store users, products, carts, orders, payments and deliveries.
- The system shall use an email service to send order confirmations.

1.9 Hardware and Software Requirements

Item	Server side	Client side
Processor	Quad-core 2.5 GHz or better	Any dual-core processor
RAM	8 GB or more	2 GB or more
Storage	100 GB SSD or more	Not significant
Network	Broadband, static IP	Internet connection
Operating system	Linux or Windows Server	Windows, Linux, macOS, Android or iOS
Software	Web server, backend framework, database server, payment gateway API, SMTP service	Modern web browser

1.10 Assumptions and Constraints

Assumptions

- Users have internet access and a modern browser.
- Users have a valid email address and phone number.
- Product information and stock are kept up to date by the administrator.
- The payment gateway and email service are available.

Constraints

- Development must be completed within the semester timeline by a team of five students.
- The system must use open-source or free-tier tools.
- Payments must be processed only through the third-party gateway, for security compliance.
- Delivery tracking depends on status updates entered by the administrator.

2 Software Design Document (SDD)

2.1 System Architecture

The system follows a layered three-tier architecture. The browser sends a request to the REST API; the relevant service applies the business rules, reads or writes the database and returns a response. The payment service calls the external gateway and updates the order based on the result.

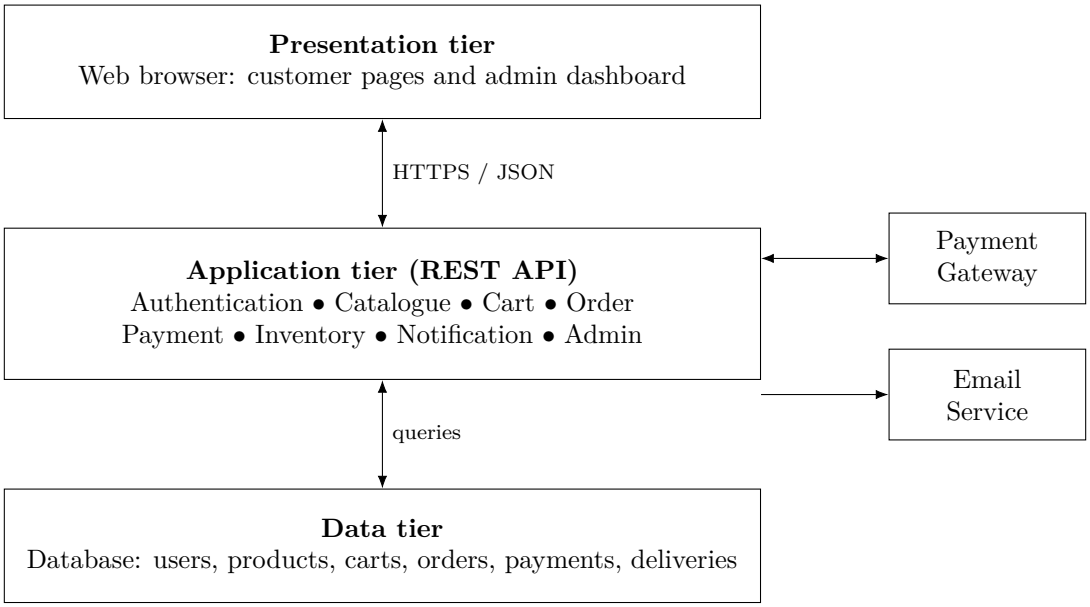


Figure 1: Three-tier architecture of the Online Shopping System

2.2 Module / Component Description

Module	Description	Main functions
User Management	Handles accounts and sessions	Register, login, logout, profile and address management, block or unblock
Product Catalogue	Manages products and categories	Add, edit, delete, list, search, filter, view details
Cart	Holds items a customer intends to buy	Add item, update quantity, remove item, calculate total
Order Management	Creates and manages orders	Place order, cancel order, view history, update status
Payment	Processes payments	Initiate payment, verify gateway response, record payment, COD handling
Delivery Tracking	Tracks shipments	Update delivery status, show tracking timeline
Inventory	Maintains stock	Reduce stock on order, restore on cancellation, low-stock check
Review	Ratings and feedback	Add and display reviews
Notification	Sends messages	Order confirmation and status emails
Admin / Reports	Administrative tools	Dashboard, sales reports, user control

2.3 Database Design

Not required by the assignment.

2.4 Class Diagram

2.5 Use Case Diagram

Actors: Customer, Administrator and Payment Gateway.

2.6 Sequence Diagrams

Place order and pay.

Login.

2.7 State Diagram

State diagram of the Order object.

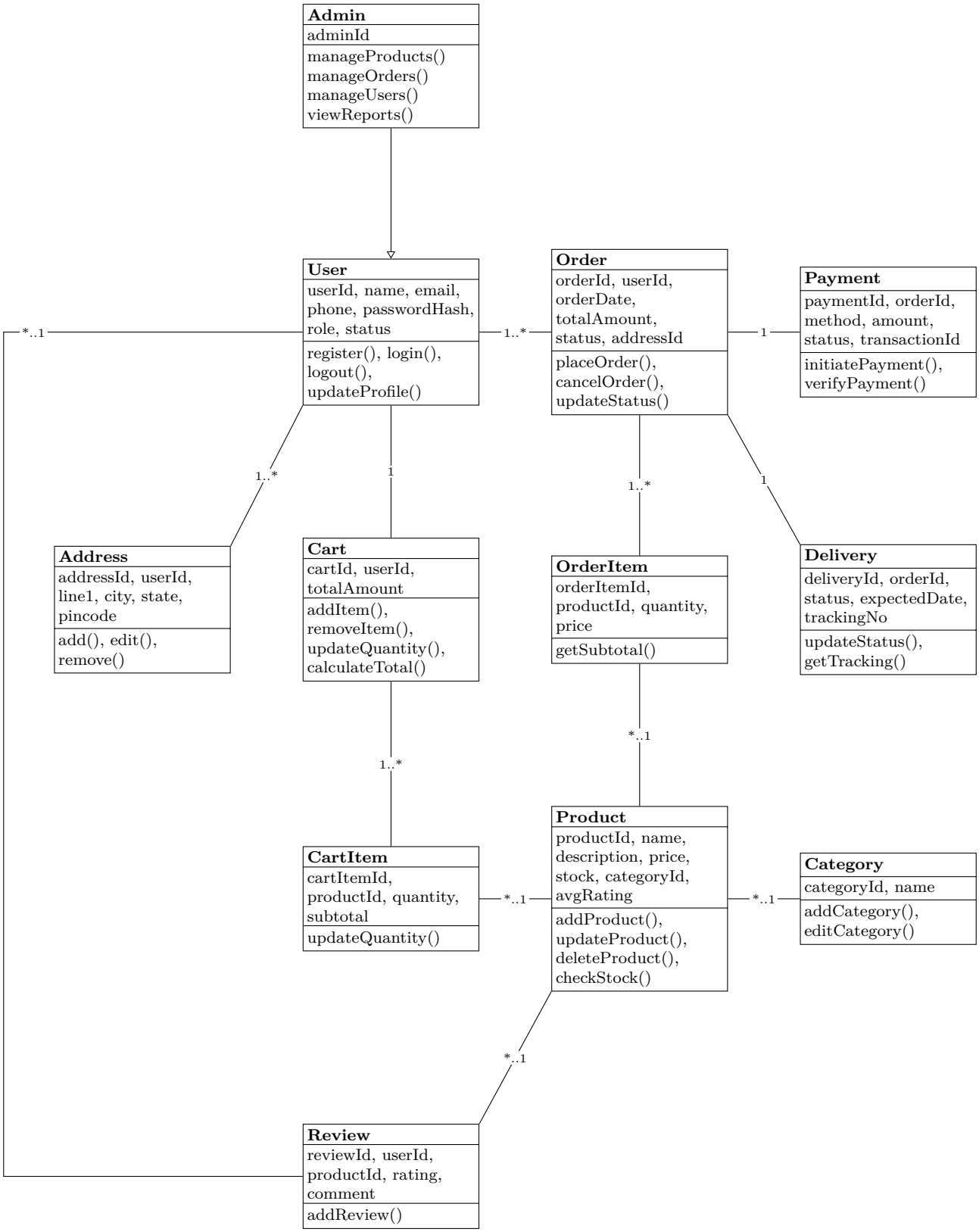


Figure 2: Class diagram (Admin specialises User; multiplicities shown on associations)

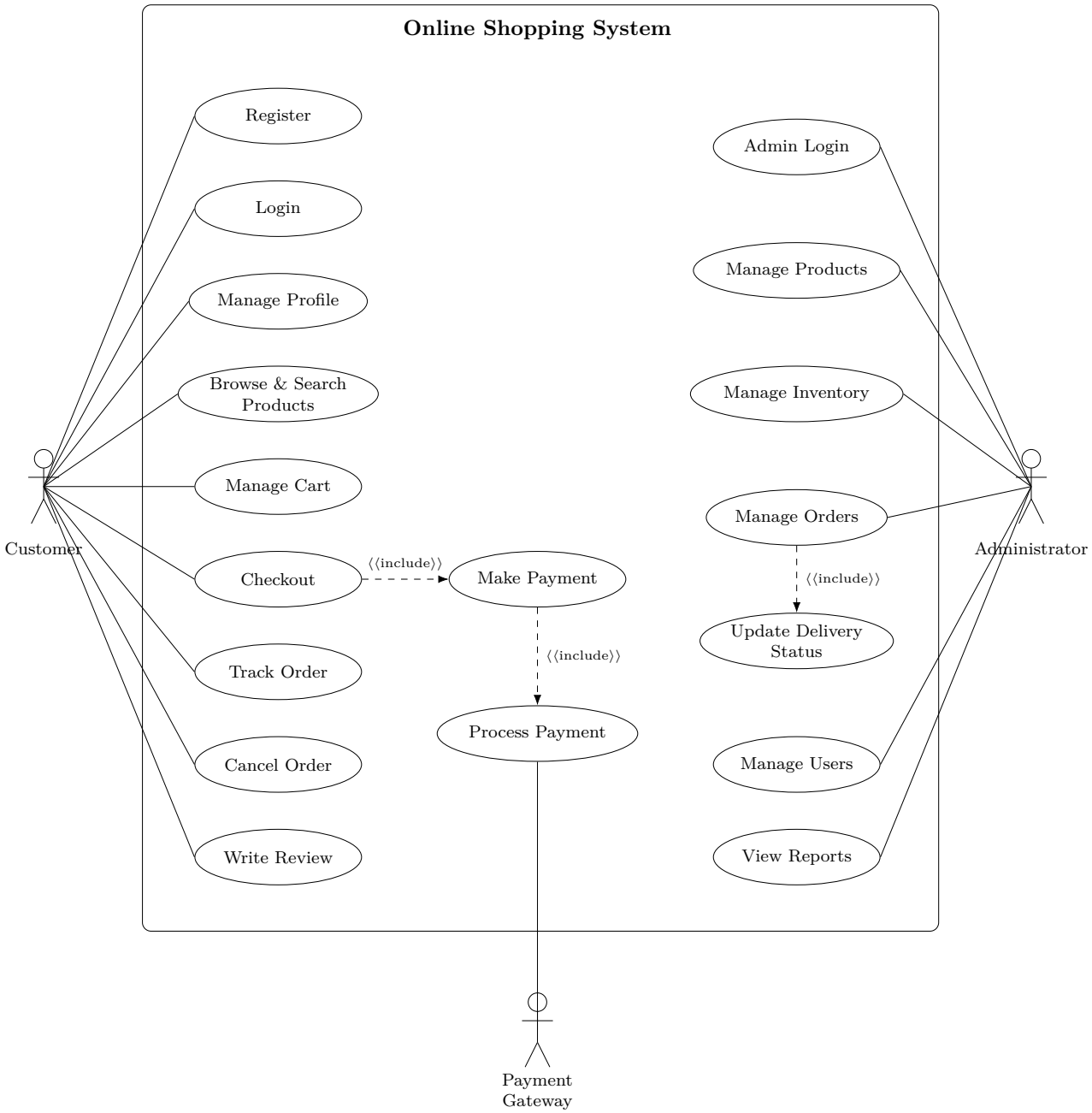


Figure 3: Use case diagram

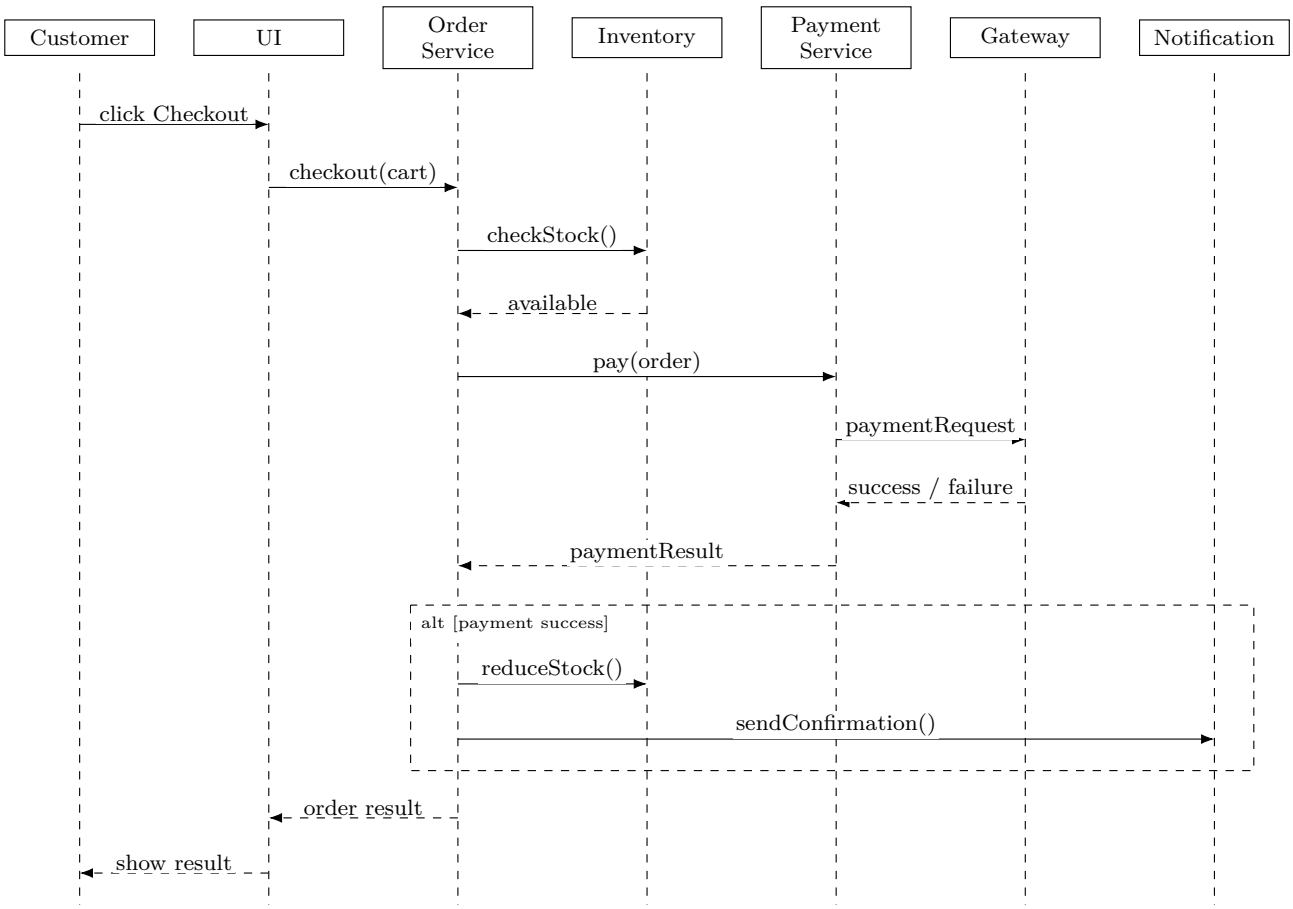


Figure 4: Sequence diagram: place order and pay

3 Software Test Plan and Report

3.1 Testing Objectives

- Verify that every functional requirement (FR1 to FR22) works as specified.
- Find and remove defects before release.
- Confirm that security, performance and usability requirements are met.
- Make sure the modules work correctly together and as a complete system.

3.2 Testing Scope

In scope: registration and login, catalogue and search, cart, checkout, payment (using the gateway sandbox), order tracking and cancellation, reviews, and all admin functions.

Out of scope: internals of the real payment gateway and email provider, courier operations, and native mobile applications.

3.3 Testing Strategy

Item	Detail
Approach	Bottom-up: unit tests first, then integration, system and acceptance tests.
Methods	Black-box (equivalence partitioning, boundary value analysis) and white-box (statement and branch coverage for core logic).

Item	Detail
Environment	Test server with a test database and payment gateway sandbox; Chrome and Firefox.
Entry criteria	Module code complete and builds without errors.
Exit criteria	All high-priority test cases pass and no critical or major defect is open.
Tools	Unit test framework of the chosen language, Postman for API testing, browser for manual testing.

3.4 Unit Testing

Each function or class is tested alone, with dependencies mocked. Targets: password validation, price and cart total calculation, stock check, order status transition rules, payment response handling.

3.5 Integration Testing

Modules are combined step by step: Cart with Product Catalogue; Order with Inventory; Order with Payment; Payment with the gateway sandbox; Order with Notification and Delivery Tracking.

3.6 System Testing

The complete application is tested against the SRS: end-to-end purchase flow, admin flow, security checks (unauthorised access, SQL injection, session handling), performance with 500 simulated users, and browser compatibility.

3.7 Acceptance Testing

Performed with sample users acting as customers and an administrator in realistic scenarios. The system is accepted if the users complete the main tasks (search, buy, pay, track, manage products and orders) without help and with no blocking defect.

3.8 Test Cases, Expected Results, Actual Results and Pass/Fail

The Actual Result and Status columns are filled in when each test is executed on the built system.

Unit tests

ID	Test case	Input	Expected result	Actual result	Status
UT-01	Password validation	“abc” (too short)	Rejected with message		
UT-02	Password validation	“Shop@1234”	Accepted		
UT-03	Cart total	$2 \times 500 + 1 \times 250$	Total = 1250		
UT-04	Stock check	Quantity 5, stock 3	Error: insufficient stock		
UT-05	Order status rule	Delivered, then Cancel	Transition rejected		
UT-06	Email format check	“user@”	Rejected		

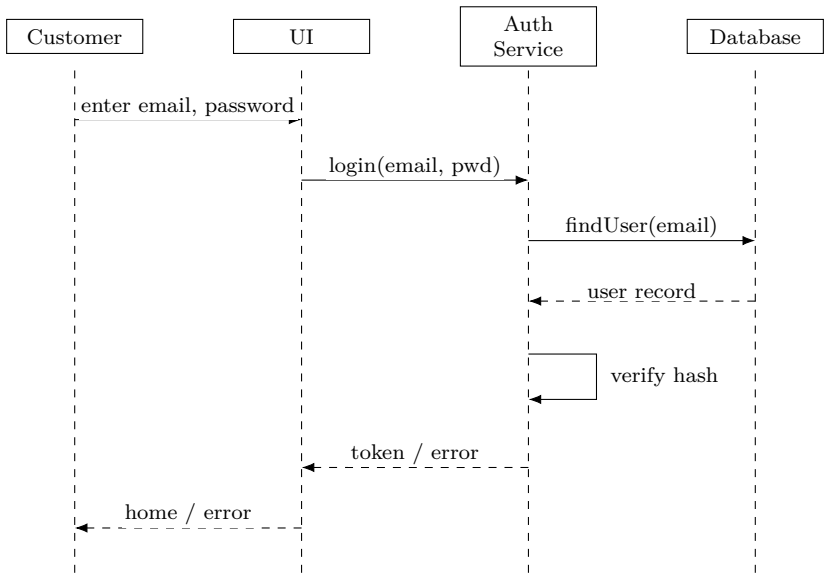


Figure 5: Sequence diagram: login

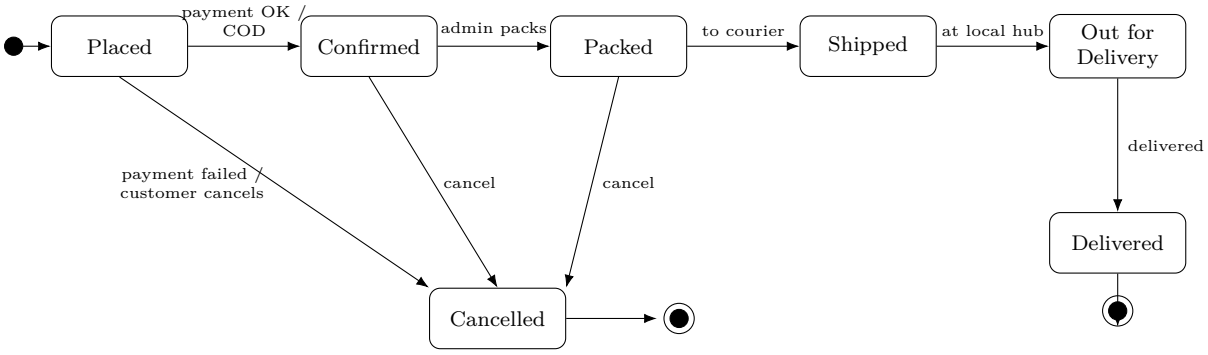


Figure 6: State diagram of an Order

Integration tests

ID	Test case	Input	Expected result	Actual result	Status
IT-01	Add to cart from catalogue	Select product, quantity 1	Item appears in cart with correct price		
IT-02	Order reduces stock	Order 2 units (stock 10)	Stock becomes 8		
IT-03	Successful payment updates order	Sandbox success response	Payment Paid, order Confirmed		
IT-04	Failed payment	Sandbox failure response	Order stays Payment Pending, stock unchanged		
IT-05	Cancellation restores stock	Cancel confirmed order of 2 units	Stock increases by 2		
IT-06	Confirmation email	Order confirmed	Email sent with order ID		

System tests

ID	Test case	Input	Expected result	Actual result	Status
ST-01	End-to-end purchase	Register, login, search, add, checkout, pay	Order created, confirmation shown		
ST-02	Login with wrong password	Valid email, wrong password	Error message, no session		
ST-03	Search with no match	Keyword “zzzzzz”	“No products found” message		
ST-04	Admin adds product	Valid product details	Product visible in catalogue		
ST-05	Customer opens admin page	Direct URL	Access denied		
ST-06	SQL injection attempt	`' OR 1=1 -` in login	Input rejected, no login		
ST-07	Load test	500 concurrent users	Pages load within 3 seconds		
ST-08	Order tracking	Admin sets status Shipped	Customer sees Shipped on tracking page		
ST-09	Browser compatibility	Chrome, Firefox, Edge	Same behaviour in all		

Acceptance tests

ID	Test case	Input	Expected result	Actual result	Status
AT-01	New customer buys a product	Sample user follows task sheet	Completes purchase without help		
AT-02	Customer tracks and cancels order	Sample user	Status visible; cancellation works before shipping		
AT-03	Admin processes an order	Sample admin	Updates status through to Delivered		
AT-04	Admin manages catalogue and stock	Sample admin	Product and stock changes show on the site		

3.9 Defect Reporting

Severity levels: Critical (system unusable or data loss), Major (key feature fails), Minor (small functional issue), Trivial (cosmetic).

Defect lifecycle: New, Assigned, Fixed, Retested, Closed (or Reopened).

Defect ID	Related test	Description	Severity	Steps to reproduce	Status	Assigned to
D-01						
D-02						
D-03						

3.10 Test Summary

Level	Total	Passed	Failed	Pass %
Unit	6			
Integration	6			
System	9			
Acceptance	4			
Total	**25**			